

KILLARA HIGH SCHOOL
YEAR 12

EXTENSION MATHEMATICS
APRIL 2002

Answer all questions and show all necessary working. Time allowed: 2 hours.

QUESTION 1 (14 marks)

- | | | | |
|----|--|--|---|
| a) | Differentiate | (i) $y = e^x \cos x$ | 1 |
| | | (ii) $y = e^{\sin x}$ | 2 |
| | | (iii) $y = \frac{\log_e x}{x}$ | 2 |
| | | | |
| b) | Find the values of m for which the equation $y = e^m + 4e^{-m}$ has a stationary point and show this is a minimum turning point. | | 3 |
| | | | |
| c) | Evaluate | (i) $\int_0^1 \frac{e^x}{1+e^x} dx$ | 2 |
| | | (ii) $\int_0^1 \frac{\sin x}{3 - \cos x} dx$ correct to 2 decimal places | 2 |
| | | (iii) $\int_1^2 \frac{x^3 + 1}{x} dx$ exactly | 2 |

QUESTION 2 (14 marks)

- a) (i) Show that $\sin^4 x = \frac{3}{8} - \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x$ 3
- (ii) Hence or otherwise find the exact value of $\int_0^{\frac{\pi}{4}} \sin^4 x \, dx$ 2
- b) (i) Sketch the curve $y = 2 - \sin x$ for $0 \leq x \leq \pi$ 2
- (ii) Find the area between this curve and the X axis and ordinates $x = 0$ and $x = \pi$. 2
- (iii) If this area is rotated about the x axis, find the volume of the solid so formed (leaving your answer in terms of π). 2
- c) Show that the locus of the point of intersection of the tangent at the extremities of the focal chord of the parabola $x^2 = 4y$ is a straight line $y + 1 = 0$. 3

QUESTION 3 (13 marks)

- a) Given $f(x) = \sqrt{1 - x^2}$, evaluate $f^{-1}(\frac{3}{5})$ 2
- b) Solve $\log_3(9x - 2) = 2 + 2 \log_3 x$ 2
- c) A particle moves along a straight line and its displacement x (metres) from a fixed point O is given by $x = 2 \ln(t + 1)$
- where t = time in seconds.
- (i) What is the initial position of the particle? 1
- (ii) What is its initial velocity? 2
- (iii) What is its initial acceleration? 2
- (iv) When does the particle have a velocity of 1.5 ms^{-1} ? 2
- (v) Show that the particle never comes to rest. 2

QUESTION 4 (12 marks)

- a) A southerly wind is blowing at a steady speed of 36 km/hr. Clouds are moving horizontally with the wind and heavy rain drops are falling to the ground 200m below.

(Assume no air resistance and take $g = 10ms^{-2}$)

- (i) Find the time taken for a rain drop to reach the ground. 2

- (ii) Find (a) the speed, and 2

- (b) the angle 2

at which the drop hits the horizontal ground.

- (iii) At what angle does a drop hit the ground when the wind speed is doubled? 2

- b) A balloon is expanding so that its surface area is increasing at a rate of $0.05cm^2/sec$. Calculate the rate of change of volume when its diameter is 8cm. 4

QUESTION 5 (12 marks)

- (a) Sketch the graphs of

- (i) $y = 2 \tan^{-1}x$ 3

- and (ii) $y = \cos^{-1} \frac{x}{2}$ 3

State their largest possible domain and range.

- (b) What is the value of

- (i) $\tan^{-1}(0)$ 1

- (ii) $\cos^{-1}(-\frac{1}{2})$ 1

- (c) What is the angle of inclination of the tangent to the curve $y = \sin^{-1}x$ at the origin? 2

- (d) What is the equation of the normal to the curve at this point? 2

END OF PAPER

Question 1 (14)

A. i $\frac{dy}{dx} = e^x (\cos x - \sin x)$

ii $\frac{dy}{dx} = \cos x \cdot e^{\sin x}$

iii $\frac{dy}{dx} = \frac{1 - \log x}{x^2}$

B. $y = e^M + 4e^{-M}$

$\frac{dy}{dx} = e^M - 4e^{-M}$

$= 0$ when $e^M = \frac{4}{e^M}$

$M = \log 2$

$\frac{d^2y}{dx^2} = e^M + 4e^{-M}$

> 0 at $M = \log 2$

$\therefore$ Minimum

C. i $\int_0^1 \frac{e^x}{1+e^x} dx = [\log(1+e^x)]_0^1$

$= \log(1+e) - \log(1+1)$

ii $\int_0^1 \frac{\sin x}{3 - \cos x} dx = [\log(3 - \cos x)]_0^1$

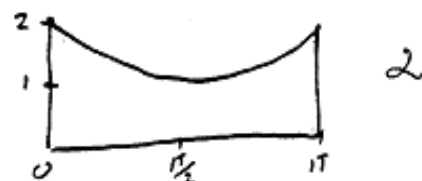
$= [\log(3 - \cos x)]_0^1$

$= \ln(3 - \cos 1) - \ln 2$

iii $\int_1^2 \frac{x^3+1}{x} = \left[\frac{x^3}{3} + \ln x \right]_1^2$

$= \frac{7}{3} + \ln 2$

Q2B:



ii $A = \int_0^\pi 2 - 2 \sin x dx$
 $= [2x + 2 \cos x]_0^\pi$
 $= 2\pi - 2$ units²

iii $V = \pi \int_0^\pi y^2 dx$
 $= \pi \int_0^\pi (2 - 2 \sin x)^2 dx$
 $= \pi \int_0^\pi 4 - 4 \sin 2x + \sin^2 2x$
 $= \pi \left[\frac{4x}{2} + \frac{1}{2} \cos 2x - \frac{1}{4} \sin 2x \right]_0^\pi$
 $= \pi \left[\frac{4\pi}{2} - \frac{1}{4} \right]$
 $= \frac{9\pi^2}{2} - 8\pi$ units³

3 2C. $x^2 = 4y$, $A=1$. Let P & Q be pts

Eqns tangent P: $y = P_1x - P_1^2$

$Q = y = Q_1x - Q_1^2$

$(P-Q)x = (P^2 - Q^2)$

$x = P+Q$

$y = PQ$

Focal Chord $PQ = -1$

$x = \frac{P^2 - 1}{P}$, $y = -1$

$\therefore y+1=0$

Question 2 (14)

A i $\sin^4 x = \sin^2 x \cdot \sin^2 x$

$= (1 - \cos^2 x)(1 - \cos^2 x)$

$= \frac{1}{4}(1 - \cos 2x)^2$

$= \frac{1}{4}(1 - 2\cos 2x + \cos^2 2x)$

$= \frac{1}{4} - \frac{1}{2}\cos 2x + \frac{1}{4}(1 + \cos 4x)$

$= \frac{1}{4} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x + \frac{1}{8}$

$= \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$

ii $\int_0^{\pi/4} \sin^4 x dx = \left[\frac{3x}{8} - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x \right]_0^{\pi/4}$

$= \frac{3\pi}{32}$

Question 3

2 A. $f(x) = \sqrt{1-x^2}$, $f'(x) = \frac{-x}{\sqrt{1-x^2}}$

At $x = \frac{3}{4}$

$f'(x) = -\frac{3}{4}$

2 B. $\log_3(9x-2) = 2 + 2\log_3 x$

$\therefore \log_3 \frac{9x-2}{x^2} = 2$

$\frac{9x-2}{x^2} = 9$

$9x^2 - 9x + 2 = 0$

$x = \frac{1}{3}, \frac{2}{3}$

3 CONT)

c. $x = 2 \ln(x+1)$

$$v = \frac{2}{x+1}$$

$$a = \frac{-2}{(x+1)^2}$$

1. A. At $t=0$, $x=0$

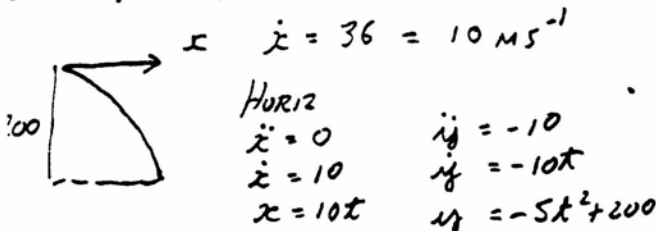
2. B. $t=0$, $v = 2 \text{ ms}^{-1}$

3. C. $t=0$, $a = -2 \text{ ms}^{-2}$

4. D) $1.5t = 2 - 1.5$
 $t = \frac{1}{3} \text{ sec.}$

5. E) For rest, $\frac{2}{x+1} = 0$ - IMPOSSIBLE

Q. 4A)



1. At ground $y=0$

$$\therefore -5t^2 + 20 = 0$$

$$t^2 = 40$$

$$t = 2\sqrt{10} \text{ SECS.}$$

2. At ground $\dot{x} = 10 \text{ ms}^{-1}$

$$\dot{y} = -20\sqrt{10}$$

$$v = \sqrt{\dot{x}^2 + \dot{y}^2}$$

$$= \sqrt{10^2 + (20\sqrt{10})^2}$$

$$= \sqrt{4100}$$

$$\div 64.03 \text{ ms}^{-1}$$

Let α = ANGLE



$$\tan \alpha = \frac{20\sqrt{10}}{10} = 2\sqrt{10}$$

$$\alpha = 81^\circ (\text{APP})$$

c. If $\dot{x} = 20 \text{ ms}^{-1}$ $\tan \alpha = \frac{20\sqrt{10}}{20} = \sqrt{10}$

$$\alpha = 72.27^\circ$$

4B) $\frac{dA}{dt} = 0.05$ $A = 4\pi R^2$ ✓

$$\frac{dA}{dR} = 8\pi R$$

$$\frac{dA}{dt} = \frac{dA}{dR} \times \frac{dR}{dt}$$

$$0.05 = 8\pi R \times \frac{dR}{dt}$$

$$\frac{dR}{dt} = \frac{0.05}{8\pi R}$$
 ✓

Now $\frac{dV}{dt} = \frac{dV}{dR} \times \frac{dR}{dt}$ ✓

$$= \frac{4\pi R^2}{3} \times \frac{0.05}{8\pi R}$$

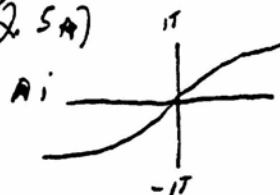
$$= \frac{R}{40}$$

WHEN $D=8$, $R=4$,

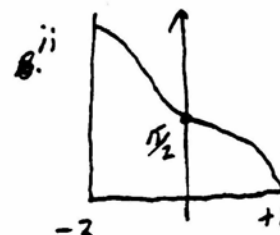
$$\frac{dV}{dt} = \frac{4}{40} = \frac{1}{10}$$
 ✓

$\therefore$ VOL INCR. AT $0.1 \text{ cm}^3 \text{ s}^{-1}$

Q. 5A)



Domain: $-\pi < x < \pi$
Range: $-\pi < y < \pi$



Domain: $-2 \leq x \leq 2$
Range: $0 \leq y \leq \pi$

B) i. $\tan^{-1}(0) = 0$

ii. $\cos^{-1}(-\frac{1}{2}) = \frac{2\pi}{3}$

c. $y = \sin^{-1}x$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

At $x=0$, $\frac{dy}{dx} = 1$

$\therefore$ ANGLE = $\frac{\pi}{4}$

D. GRAD OF NORMAL = -1.

$y = -x$ EQN.